EMANI HITESH REDDY

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Projects

Multi format QR-Code Generator — Github repo

- Developed a cross functional QR Code Generator in C++ supporting Website, Plain Text, Wi-Fi, Email, Phone, Contact (vCard), and UPI Payment QR generation.
- Developed an XML-based SVG generation module to produce high-quality, scalable QR code outputs.
- Designed a persistent history logging system using C++ file handling.

Smart Agriculture System — Github repo

- Developed a smart agriculture system using ESP32, sensors, and a web-based monitoring interface
- Implemented embedded firmware for sensor data acquisition and real-time status updates
- Integrated machine learning model for irrigation prediction and decision support.

4-Bit Binary Calculator — Github repo

- Designed, implemented, and validated embedded firmware for an LPC1768 (ARM Cortex-M3) based arithmetic processing system with real-time GPIO and display peripheral interfacing.
- Integrated multiplexed seven-segment displays and binary LED indicators to provide synchronized hexadecimal and binary output visualization.

Internship

Research Intern | SRM University, AP — May 2025 – July 2025

- Developed a HDL code for a real-time unmanned Parking System using verilog.
- Developed RTL modules for parking slot management, vehicle authentication, and real-time occupancy tracking using combinational and sequential logic.

VLSI Design Intern | SkillOrbit X Kshitij, IIT Kharagpur (Online) — Jun 2026 – Present

- Learned Verilog HDL fundamentals including developement of source and testbench codes.
- Designed and simulated RTL modules using Verilog HDL.
- Gained hands-on experience in HDL programming

Education

2023 - 2027 Bachelor of Technology in ECE, SRM University, AP (CGPA: 7.80)

Skills

Programming Languages	:	Embedded C, C++, C-Language
Embedded Systems	:	Firmware Development, Embedded Systems, Hardware-Software Integration,
Microcontrollers	:	ESP32, Arduino Uno, ARM Cortex-M
HDL & FPGA	:	Verilog, RTL Design
Development Tools	:	Arduino IDE, Xilinx Vivado, Cadence
Version Control	:	Git, GitHub

Certifiactions

Maven Silicon | Verilog HDL - Hands on — Certificate